THINK, VERIFY, STOP

A Small-Scale Study of Adaptive Test-Time Compute in Open Reasoning Models

Technical Project Report

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ABSTRACT

This pilot study evaluates whether additional and adaptively allocated inference-time computation improves mathematical reasoning in an open language model. Using Qwen3.5-4B in four-bit precision, three strategies were tested on a reproducibly sampled set of 20 GSM8K problems: strict non-thinking generation, fixed 512-token reasoning, and a disagreement-triggered adaptive policy. Fixed reasoning achieved 80% accuracy, compared with 15% for strict non-thinking. The adaptive policy reached 60% while consuming more tokens than fixed reasoning. The central negative result is that agreement between two short samples was not a dependable confidence signal: five of eight early agreements were incorrect. The findings support test-time reasoning as a powerful capability lever while showing that adaptive stopping requires independent, calibrated verification rather than answer agreement alone.

1. Introduction

Modern reasoning models can spend additional computation after receiving a problem, rather than relying only on capability acquired during training. This test-time compute paradigm includes longer reasoning traces, sampling multiple candidate solutions, search, verification, revision, and adaptive allocation of effort. The central engineering question is not simply whether more computation helps, but how it should be allocated so that difficult tasks receive additional effort without imposing unnecessary cost on easy tasks.

Prior research established that chain-of-thought prompting can improve multi-step reasoning (Wei et al., 2022), while self-consistency can further improve performance by sampling diverse reasoning paths and aggregating their answers (Wang et al., 2022). More recent work shows that test-time computation can outperform parameter scaling on suitable tasks when computation is allocated according to problem difficulty (Snell et al., 2024). At the same time, confidence-aware adaptive sampling remains an active research area because naive consistency signals can be expensive or miscalibrated (Xiong et al., 2026).

1.1 Research Objective

This project investigates whether a simple disagreement-triggered adaptive policy can retain the accuracy benefit of explicit reasoning while using less computation than a uniform high-budget strategy.

1.2 Research Questions

1. How much does explicit inference-time reasoning improve accuracy over a strict direct-answer baseline?
2. Can agreement between independently sampled short reasoning attempts serve as a reliable confidence signal?
3. Can a disagreement-triggered adaptive policy improve the accuracy-compute trade-off relative to fixed reasoning?

1.3 Hypotheses

- H1: Fixed inference-time reasoning will substantially improve accuracy over strict non-thinking generation.
- H2: Independently sampled short attempts will sometimes agree on the same incorrect answer because their errors are correlated.
- H3: A naive agreement-based policy may fail to outperform fixed reasoning unless stopping decisions use stronger verification signals.

2. Related Work

GSM8K was introduced as a linguistically diverse benchmark of 8,500 grade-school mathematical word problems requiring multi-step reasoning (Cobbe et al., 2021). The benchmark has been widely used to study generators, verifiers, chain-of-thought prompting, and sampling-based inference.

Self-consistency replaces a single reasoning trajectory with multiple sampled paths and selects the most consistent final answer. Although effective, it increases inference cost and assumes that the most frequent answer is usually correct. The present pilot directly tests a lightweight version of that assumption using only two short samples and an escalation rule.

3. Methodology

3.1 Model and Runtime

- Model: Qwen/Qwen3.5-4B, an open model with thinking and non-thinking modes.
- Quantization: four-bit NF4 using bitsandbytes 0.50.0.
- Compute: NVIDIA Tesla T4 GPU in Google Colab.
- Software: Python 3.12.13, PyTorch 2.11.0, Transformers 5.13.1, Datasets 4.0.0.
- Reasoning finalization: a deterministic non-thinking pass extracted a boxed numeric answer from each reasoning trace.

Qwen3.5-4B operates in thinking mode by default and can be switched to non-thinking mode, enabling a controlled comparison within the same model family (Qwen Team, 2026). The four-bit configuration was selected to fit the model on a freely available T4 GPU. Quantization may affect absolute performance and is treated as a limitation.

3.2 Benchmark and Sampling

The study used 20 examples drawn from the 1,319-item GSM8K test split. Sampling used seed 2026, and the selected dataset indices were recorded in the public notebook to support exact reproduction. Ground-truth answers were extracted from the final GSM8K answer marker and compared numerically with model outputs.

3.3 Inference Strategies

Strategy A: Strict Non-Thinking

The model was instructed to return only a boxed numeric answer, with thinking disabled. A maximum of 128 new tokens was allowed, although naturally completed responses averaged fewer than 10 generated tokens.

Strategy B: Fixed 512-Token Thinking

Every problem received one sampled thinking pass with a maximum of 512 tokens, followed by a deterministic answer-finalization pass. The same reasoning allocation was applied to every question.

Strategy C: Naive Adaptive Agreement

Each problem first received two independently seeded 128-token reasoning attempts. If the extracted answers agreed numerically, the policy stopped early. If they disagreed, it escalated to a 512-token reasoning pass. All attempts included deterministic finalization.

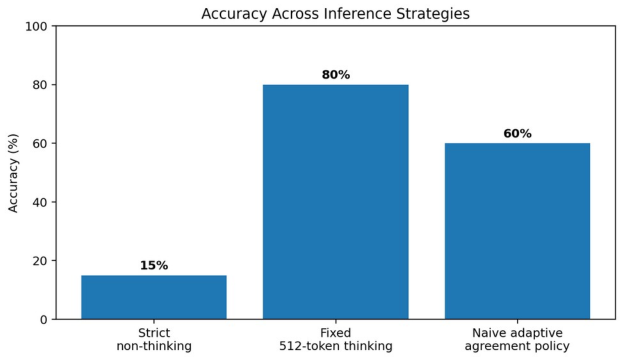
3.4 Evaluation Metrics

- Exact numerical accuracy.
- Average generated tokens per problem.
- Average generation time per problem.
- Early agreement and escalation rates.
- Correctness of early agreements.

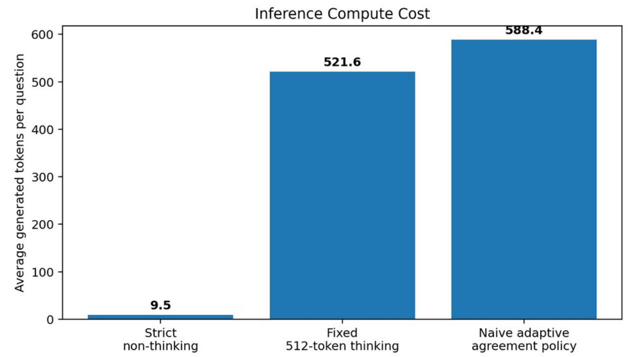
4. Results

Strategy	Accuracy	Correct	Avg. tokens	Avg. time	95% Wilson CI
Strict non-thinking	15%	3/20	9.5	1.45 s	5.2% to 36.0%
Fixed 512-token thinking	80%	16/20	521.6	47.00 s	58.4% to 91.9%
Naive adaptive agreement policy	60%	12/20	588.4	54.27 s	38.7% to 78.1%

Table 1. *Pilot results across three inference strategies. Confidence intervals are descriptive only because the sample contains 20 questions.*



Accuracy comparison



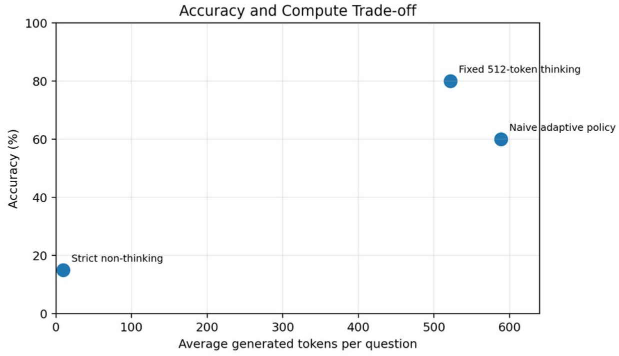
Average token cost

4.1 Primary Finding

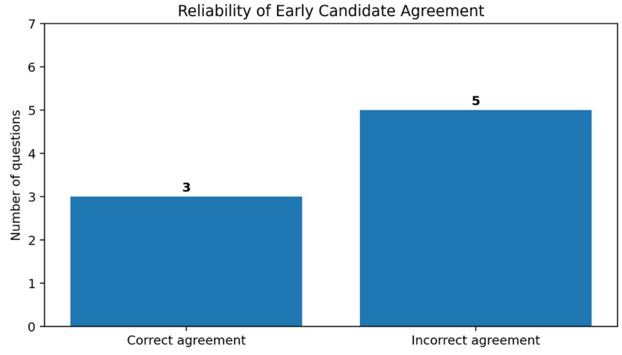
Fixed 512-token reasoning improved accuracy by 65 percentage points over strict non-thinking generation, from 15% to 80%. The cost was substantial: average generated tokens increased from 9.5 to 521.6, and average runtime increased from 1.45 seconds to 47.00 seconds per question.

The naive adaptive policy did not provide an efficiency gain. It achieved 60% accuracy, averaged 588.4 generated tokens, and took 54.27 seconds per question. Relative to fixed reasoning, it used approximately 12.8% more generated tokens while answering four fewer questions correctly.

5. Discussion



Accuracy-compute trade-off



Early-agreement reliability

5.1 Why the Adaptive Policy Failed

The adaptive policy stopped early on eight questions. Only three of those agreements were correct, producing an early-agreement precision of 37.5%. The remaining five cases demonstrate correlated error: both short trajectories converged on the same wrong answer. Answer agreement therefore measured consistency, not correctness.

For the 12 escalated questions, nine were answered correctly after the longer reasoning pass. This indicates that disagreement was a useful warning signal, but agreement was not sufficient evidence for stopping. The result is consistent with the broader motivation for verifier-guided and confidence-aware test-time compute: adaptive systems need signals that evaluate reasoning quality, not merely repetition.

5.2 Research and Engineering Implications

- A fixed reasoning budget can be a strong baseline and should not be assumed inferior to a more complex adaptive controller.
- Multiple samples from the same model are not statistically independent in a practical sense; shared model biases can create correlated mistakes.
- Adaptive stopping should use independent evidence such as process verifiers, symbolic arithmetic checks, calibrated token-level uncertainty, or model diversity.
- Negative results are valuable because they identify which intuitive confidence heuristics are unsafe to deploy without validation.

6. Limitations

- The sample size was 20, so the results are exploratory and no statistical significance claim is made.
- Only one model and one benchmark were evaluated.
- Four-bit quantization may change reasoning quality relative to full-precision inference.
- Runtime measurements were collected on a shared Colab T4 environment and are not hardware-independent.
- The finalizer used the same model family, so extraction errors and correlated biases remain possible.
- The adaptive policy did not include a trained verifier, external tool, or confidence calibration model.

7. Conclusion

This study demonstrates both the value and the difficulty of adaptive test-time reasoning. Explicit reasoning produced a major accuracy improvement over direct-answer generation. However, a simple adaptive controller based on agreement between two short samples was neither more accurate nor more efficient than a fixed reasoning budget.

The core lesson is that deciding when to stop is itself a reasoning and verification problem. Consistency is not equivalent to correctness, particularly when samples originate from the same model and share systematic failure modes. Future work should evaluate independent process verifiers, tool-assisted arithmetic checking, model-diverse sampling, learned confidence estimators, and larger benchmark samples.

The project contributes a transparent, reproducible pilot framework, complete raw outputs, and an openly reported negative result. It does not claim a new state-of-the-art method. Its purpose is to demonstrate rigorous experimental thinking about a frontier problem in efficient reasoning systems.

References

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Reproducibility and Integrity

The public repository includes the executed notebook, raw CSV files, figures, requirements, and a research integrity statement. The experiment was developed with AI assistance, then run, inspected, tested, and validated by the author, who accepts responsibility for the implementation and reported results.